

Stock Market Anomaly Detection Using Machine Learning

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1. Introduction

Financial markets are characterized by periods of stability interspersed with sudden and significant fluctuations driven by macroeconomic events, firm-specific news, and changes in investor sentiment. Detecting such abnormal market behavior is crucial for market surveillance, risk management, and investment decision-making.

This project develops a machine learning-based anomaly detection system to identify unusual trading patterns in equity markets. Using historical stock price and volume data, the model learns the normal behavior of financial time series and flags deviations that may correspond to market shocks or irregular trading activity.

The analysis focuses on a cross-section of stocks across multiple sectors, including technology, financial services, and energy, to capture diverse market dynamics.

2. Objective

The primary objective of this study is to:

- Develop a system that detects abnormal trading behavior in stock markets
- Identify periods of extreme price movements, volatility, and volume spikes
- Compare anomaly patterns across different sectors
- Demonstrate the application of machine learning in financial market surveillance

3. Data Description

The dataset consists of daily stock market data obtained using Python's yfinance library. The analysis includes the following stocks:

- Technology: Apple (AAPL), Tesla (TSLA)
- Financials: JPMorgan Chase (JPM), Goldman Sachs (GS)
- Consumer/Tech: Amazon (AMZN)
- Energy: Exxon Mobil (XOM)

The dataset spans from January 2020 to December 2025, capturing significant market events such as the COVID-19 crash and subsequent recovery.

For each stock, the following variables were used:

- Adjusted closing price
- Trading volume
- Daily high and low prices

4. Methodology

4.1 Feature Engineering

To capture market behavior effectively, several financial indicators were constructed:

- **Daily returns:** percentage change in price
- **Log returns:** continuous compounding returns
- **Rolling volatility:** 20-day standard deviation of returns
- **Average trading volume:** rolling mean of volume
- **Abnormal volume:** ratio of current volume to average volume
- **Price range:** intraday volatility (high–low relative to price)
- **Momentum (5-day):** short-term price trend

These features provide a comprehensive representation of both price dynamics and trading activity.

4.2 Anomaly Detection Model

The study employs the Isolation Forest algorithm, an unsupervised machine learning method designed for anomaly detection.

Isolation Forest works by:

- Randomly partitioning the data
- Isolating observations that differ significantly from the majority
- Assigning anomaly scores based on how easily points are separated

Observations classified as anomalous are those that deviate from learned patterns of normal market behavior.

4.3 Model Implementation

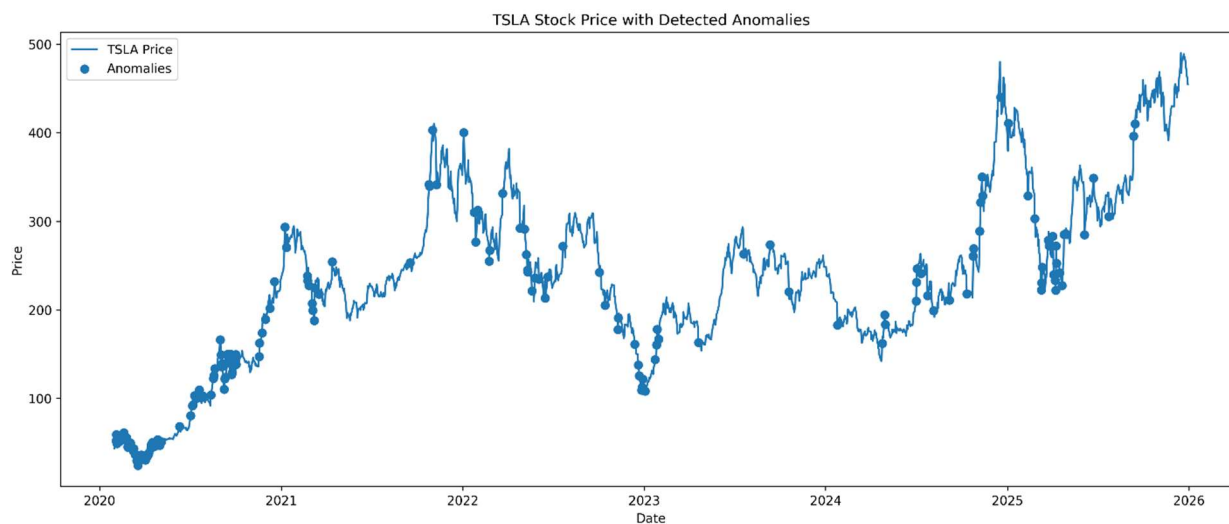
The workflow for each stock includes:

1. Data collection and cleaning
2. Feature engineering
3. Standardization of features
4. Model training using Isolation Forest
5. Identification of anomalies
6. Visualization of results

4.4 Visualization

To illustrate the effectiveness of the anomaly detection model, selected visualizations are presented for representative stocks across different levels of market volatility. These figures highlight abnormal price movements and the intensity of detected anomalies over time.

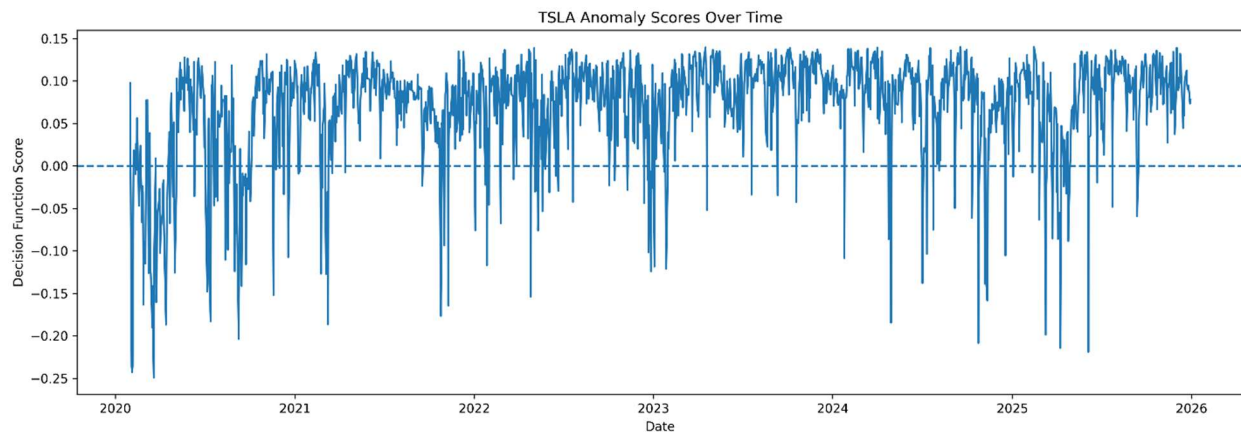
The first set of visualizations focuses on Tesla Inc. (TSLA), a highly volatile technology stock.



This figure shows the TSLA price series with highlighted anomaly points. The model detects a high concentration of anomalies during periods of significant market disruption, particularly during the COVID-19 crash in early 2020. In addition, anomalies are observed

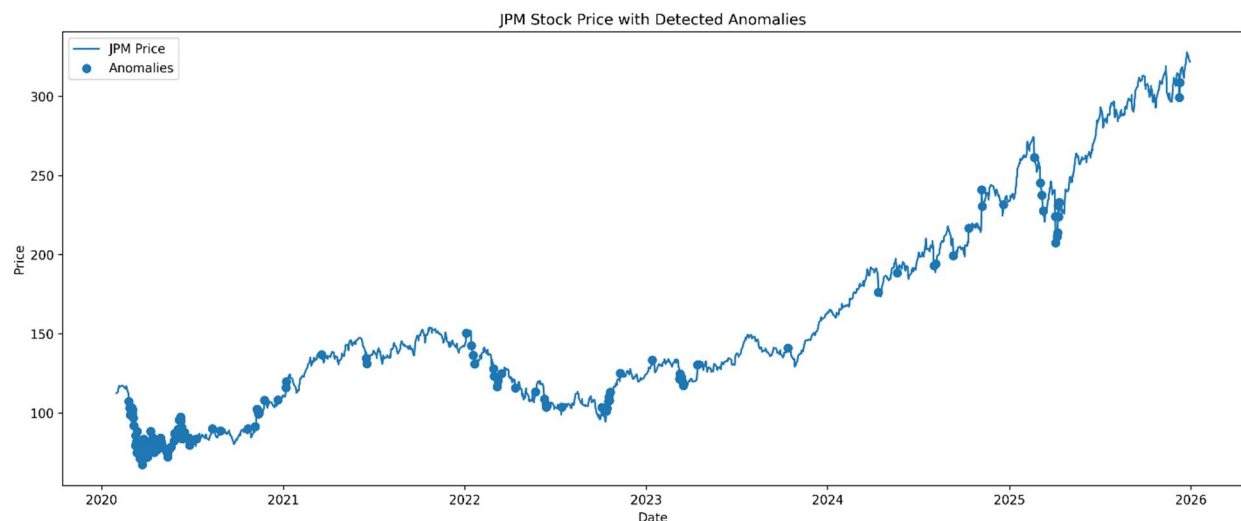
throughout later periods, reflecting Tesla’s sensitivity to market sentiment, firm-specific news, and rapid price movements. The higher frequency of anomalies compared to other stocks highlights the model’s responsiveness to volatility.

To further examine anomaly intensity, the anomaly score is presented:



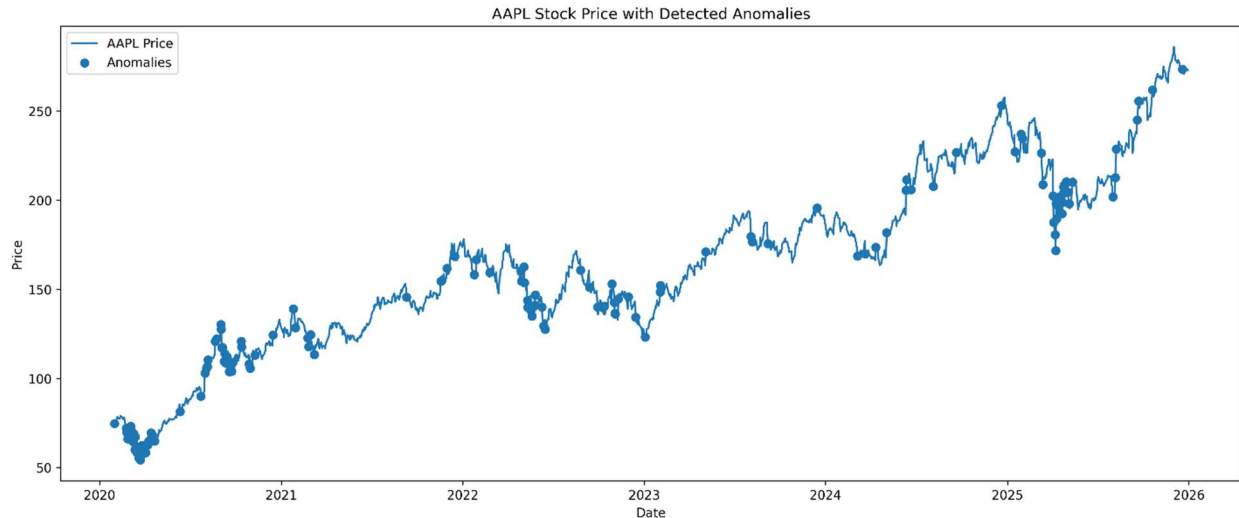
The anomaly score plot provides a continuous measure of abnormality, where lower values indicate stronger deviations from normal behavior. Sharp negative spikes correspond to extreme market events, confirming that the model assigns higher anomaly intensity to periods of substantial price changes and abnormal trading activity.

To provide a benchmark for comparison, JPMorgan Chase (JPM) is included as a relatively stable financial stock:



Compared to TSLA, JPM exhibits fewer anomaly points, indicating more stable price behavior. The anomalies that are detected are primarily concentrated during major market stress periods, such as the 2020 crash, suggesting that the model effectively distinguishes between normal fluctuations and significant disruptions in lower-volatility assets.

Finally, Apple Inc. (AAPL) is presented as a moderately volatile stock:



AAPL displays an intermediate level of anomaly frequency, with more detected events than JPM but fewer than TSLA. This reflects its position between stable financial stocks and high-growth technology stocks. The model captures both market-wide shocks and stock-specific movements, demonstrating its ability to generalize across different asset classes.

Overall, the visualizations confirm that the anomaly detection model adapts effectively to varying market conditions. It identifies extreme events in highly volatile assets while maintaining selectivity in more stable stocks, providing a robust framework for detecting abnormal trading behavior.

5. Results and Findings

5.1 Detection of Major Market Events

The model successfully identified significant periods of market stress. In particular:

- The COVID-19 market crash (March 2020) was consistently flagged across all stocks
- Extreme price declines and rebounds during this period were captured as high-intensity anomalies
- Subsequent recovery phases also exhibited abnormal behavior due to heightened volatility

5.2 Sectoral Differences in Anomalies

The analysis revealed clear differences across sectors:

- **Technology stocks (TSLA, AMZN, AAPL)** exhibited a higher frequency of anomalies
 - Reflecting greater sensitivity to market sentiment and news
 - Higher volatility and rapid price movements
- **Financial stocks (JPM, GS)** showed fewer anomalies
 - Indicating relatively more stable trading patterns
 - Lower volatility compared to high-growth tech stocks
- **Energy stock (XOM)** displayed moderate anomaly levels
 - Likely influenced by fluctuations in oil prices and macroeconomic conditions

5.3 Anomaly Intensity

By introducing an anomaly strength metric (derived from anomaly scores), the model was able to rank abnormal events by severity.

The strongest anomalies corresponded to:

- Large negative returns during market crashes
- Sharp positive rebounds following major economic announcements
- Unusually high trading volumes

5.4 Interpretation of Anomalies

The detected anomalies can be interpreted as:

- Market-wide shocks (e.g., pandemic-driven sell-offs)
- Firm-specific events (earnings surprises, news releases)
- Liquidity and volatility spikes
- Sudden shifts in investor behavior

6. Practical Applications

This anomaly detection system has several real-world applications:

- **Market Surveillance:** Identifying unusual trading activity or potential manipulation
- **Risk Management:** Detecting periods of elevated volatility and stress
- **Trading Strategies:** Highlighting potential entry/exit points during abnormal conditions
- **Regulatory Monitoring:** Supporting oversight of financial markets

The model can be extended to operate on live data, enabling real-time anomaly detection.

7. Limitations

While effective, the model has some limitations:

- The anomaly detection threshold may produce false positives
- The model does not incorporate fundamental or news-based data
- Daily frequency may miss intra-day anomalies
- Isolation Forest does not provide causal explanations for anomalies

8. Conclusion

This project demonstrates the effectiveness of machine learning in identifying abnormal market behavior. By combining financial feature engineering with an unsupervised anomaly detection model, the system successfully captures major market events and differentiates behavior across sectors.

The results highlight the importance of volatility and trading activity in understanding market dynamics and show how data-driven approaches can support financial analysis and decision-making.

9. Future Work

Future improvements could include:

- Incorporating macroeconomic indicators
- Integrating news sentiment analysis

- Using higher-frequency (intra-day) data
- Comparing multiple anomaly detection models
- Developing real-time monitoring systems